Yoonhyuk Choi (Last update: Mar. 2026)

Assistant Professor
Sookmyung Women's University
Department of Artificial Intelligence

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Research Interest

My research interests lie in *Machine Learning* and *Data Mining*, particularly in **Large Language Models**, **Graph Neural Networks**, and **Recommender Systems**.

Education

•Seoul National University — Mar. 2019 - Aug. 2023
Ph.D. & M.S., Computer Engineering — Advisor: Chong-Kwon Kim

•University of Seoul — Mar. 2013 - Feb. 2019
B.S., Computer Science — Advisor: Eui-Kyeong Hong

Work Experience

•Assistant Professor — Sep. 2025 -
Sookmyung Women's University, Seoul, South Korea
Department of AI

•Research scientist — Oct. 2024 - Aug. 2025
Samsung SDS, Seoul, South Korea
Preceding AI Lab (LLM, RAG)

•Postdoc. associate — Nov. 2023 - Sep. 2024
Arizona State University, Tempe, United States (PI: Selcuk Candan and Huan Liu)
School of Computing and Augmented Intelligence (SCAI)

•Postdoc. associate — Sep. 2023 - Nov. 2023
Korea Institute of Energy Technology, Naju, South Korea (PI: Chong-Kwon Kim)
Energy AI

•Backend engineer — Jun. 2018 - Sep. 2018
nTOPAZ, Seoul, South Korea
Back-end Engineer

Publications (Google Scholar) - C: conference, J: journal

- (C18) Sheaf Graph Neural Networks via PAC-Bayes Spectral Optimization (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Jongwook Kim, Chong-Kwon Kim
  AAAI '26

- (J17) Identifying Heterophilic Neighbors via Confidence-based Subgraph Matching for Graph Neural Networks (link)
  Yoonhyuk Choi, Chong-Kwon Kim
  Artificial Intelligence '26 (IF: 4.6)

- (J16) Beyond Binary: Improving Signed Message Passing in Graph Neural Networks for Multi-Class Graphs (link)
  Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim
  IEEE TPAMI '25 (IF: 20.8)

- (J15) Hierarchical Hyperbolic Embeddings for Review-Driven Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Chong-Kwon Kim
  IEEE Access '25 (IF: 3.6)

- (J14) Generalization of Knowledge Transfer with User Reviews for Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Chong-Kwon Kim
  IEEE Access '25 (IF: 3.6)

- (C13) Selective Blocking for Message-Passing Neural Networks on Heterophilic Graphs (link)
  Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim
  UAI '25

- (C12) Review-Based Hyperbolic Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
  WSDM '25

- (C11) Mitigating Overfitting in Graph Neural Networks via Feature and Hyperplane Perturbation (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
  WSDM '25

- (J10) Beyond Message-Passing: Generalization of Graph Neural Networks via Feature Perturbation for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
  IEEE TNNLS '24 (IF: 14.25)

- (C9) Improving the Text Convolution Mechanism with Large Language Model for ... (link)
  Yoonhyuk Choi, Fahim Tasneema Azad
  IEEE Big Data '24 (short)

- (C8) Prioritizing Potential Wetland Areas via Region-to-Region Knowledge Transfer and Adaptive Propagation (link)
  Yoonhyuk Choi, Reepal Shah, John Sabo, Selcuk Candan, Huan Liu
  IEEE Big Data '24

- (C7) Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning (link)
  Ahmet Kapkiç, Pratanu Mandal, Shu Wan, Paras Sheth, Abhinav Gorantla, **Yoonhyuk Choi**, Huan Liu, K Selçuk Candan
  CIKM '24 (benchmark)

- (C6) Universal Graph Contrastive Learning with a Novel Laplacian Perturbation (link)
  Taewook Ko, **Yoonhyuk Choi**, Chong-Kwon Kim
  UAI '23

- (J5) A spectral graph convolution for signed directed graphs via magnetic laplacian (link)
  Taewook Ko, **Yoonhyuk Choi**, Chong-Kwon Kim
  Neural Networks '23 (IF: 7.8)

- (J4) Aspect-oriented unsupervised social link inference on user trajectory data (link)
  Hyungho Byun, **Yoonhyuk Choi**, Chong-Kwon Kim
  Information Sciences '23 (IF: 8.2)

- (C3) Review-Based Domain Disentanglement without Duplicate Users or Contexts for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim
  CIKM '22

- (C2) Finding Heterophilic Neighbors via Confidence-based Subgraph Matching for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim
  CIKM '22

- (J1) Dynamic graph convolutional networks with attention mechanism for rumor detection on social media (link)
  Jiho Choi, Taewook Ko, **Yoonhyuk Choi**, Hyungho Byun, Chong-Kwon Kim
  PLOS ONE '21 (IF: 2.9)

- (arXiv) Gauge-Equivariant Graph Networks via Self-Interference Cancellation
  Yoonhyuk Choi, Jiho Choi, Jiwoo Kang
  Under Review (ICML '26)

- (arXiv) Sparse Bayesian Message Passing under Structural Uncertainty
  Yoonhyuk Choi, Jiho Choi, Chanran Kim, Yumin Lee, Hawon Shin, Yeowon Jeon, Minjeong Kim, Jiwoo Kang
  Under Review (IJCAI '26)

- (arXiv) Bias-aware Adaptive Loss for Sequential Recommendation
  Yoonhyuk Choi, Jiho Choi, Chong-Kwon Kim
  Under Review (SIGIR '26)

- (arXiv) Adaptive Branch Specialization in Spectral–Spatial Graph Neural Networks for Certified Robustness
  Yoonhyuk Choi, Chong-Kwon Kim
  Under Review (TPAMI)

- (arXiv) Learning Sheaf Graph Neural Networks with PAC-Bayesian Spectral Guarantees
  Yoonhyuk Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
  Under Review (TPAMI)

- (arXiv) Hierarchical and Uncertainty-Aware Graph Neural Networks for Heterophily and Robustness
  Yoonhyuk Choi, Chanran Kim, Yumin Lee, Chong-Kwon Kim
  Under Review (Pattern Recognition)

- (arXiv) Spectral-Radius and Uncertainty-Aware Blocking in Message-Passing Neural Networks
  Yoonhyuk Choi, Jiho Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
  Under Review (TIST)

Projects

•Enhancing Large Language Model with RAG 2025

Research project, Samsung SDS

- Personalized RAG and knowledge graph construction on Knox Systems
- Knowledge graph construction and RAG for chunk retrieval
- Tiny-LLM (e.g., Llama 3B) distillation with huge-LLM (e.g., Llama 70B)

•Can Large Language Model Improve the Text Convolution for Review-Based Recommendation? 2024

Research project, Emit Lab

- Integrated the large language model (e.g., Llama 2, GPT-4) with the text convolution algorithm
- Investigated whether applying summarization based on large language models first, instead of performing 2D convolution on the entire text, results in performance improvement
- Used online shopping mall datasets like Amazon and Walmart

•Selection Criteria and Assigned Weightage for Identifying Potential Locations Wetland 2024

Research project funded by NSF (in collaboration with Tulane University)

- Suggested knowledge transfer between different regions and adaptive propagation between grids
- Demonstrated the effectiveness of the framework through real-world scenario
- Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

•Causal Discovery of Agricultural Mgmt and Reservoir Op. Induced Water Quality Change 2023

Research project funded by NSF (in collaboration with University of Arkansas)

- Developed causal discovery algorithm for water quality improvement and reservoir management
- Considered spatial and temporal variations and validated the causal learning ability

– Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

•Tracking footprints with graph neural networks for the reduction of virus spread 2021 - 2022

Coursework project, R&D in AI industry

– Suggested spatial-temporal analysis for the next POI prediction to reduce virus spread
– Selected as social contributing project
– Used datasets are Coronamap of South Korea, Gowalla for POI prediction

•Personalized recommendation based on the user's purchasing histories and social network 2020

Industry project funded by Samsung Research

– Introduced time series analysis of users' purchasing history for personalized advertising
– Applied graph neural networks with binary recommendation techniques
– Used customer datasets provided by Samsung Research

•Next POI prediction based on user movements collected through large-scale sensors 2019

Research project funded by Samsung Electronics

– Recommending the next place based on where students visited within Seoul National University
– Developed energy-saving and effective multi-hop transmission technologies for sensor
– Collected datasets by attaching special stickers to participants

Extracurricular Activities

•Program Committee 2026

NeurIPS / ICLR / IJCAI / ICML / AAAI / KDD (Aug. track) / WSDM / TheWebConf

•Program Committee 2025

ICLR / IJCAI / ICML / KDD (Feb. track) / CIKM / WSDM / TheWebConf

•Program Committee 2024

LoG / CIKM / MM

•Invited Talk (N-EWN Partner Symposium) 2024

Titled Identifying Potential Sites for Wetlands, St. Augustine in Florida

•Reviewer 2023

Journal of IEEE Multimedia / Plos one

•Research Assistant (RA), Graduate Mar. 2019 - Jun. 2021

Funded by Samsung Research

•Teaching Assistant (TA) Mar. 2020 - Jun. 2020

Topic: Social Network Analysis and Anomaly Detection (Advisor: Chong-Kwon Kim)

•Research Assistant (RA), Undergrad Jun. 2017 - Sep. 2017

Distributed Computing Lab (Supervisor: Jin-Suk Kim)

Awards & Grants

•Best Ph.D. Dissertation Award 2023

Seoul National University

•Overseas Short-term Training Scholarship 2023

Chonnam National University

•BK21 Colloquium Graduate Student Fellowship 2023

Seoul National University

•**BK21 Star Student Researcher Fellowship** 2023
Seoul National University

•**SIGIR Travel Awards** 2022
For ACM Student Authors with Accepted Long Paper

•**BK21 Scholarship** 2020
(Graduate) Seoul National University

•**Merit-based Scholarship** 2018 - 2019
(Undergrad) University of Seoul

Skills

•**Languages**: Python, C, HTML/CSS

•**Tools / Frameworks**: PyTorch, torch-geometric, Scikit-learn, Git, Django, AWS, LaTex

References

•**ChongKwon Kim**: ckim@kentech.ac.kr

•**TaeKyung Kwon**: tkkwon98@gmail.com

•**U Kang**: ukang@snu.ac.kr